

Frank — AMC Upper Primary **BALANCED PRIZE TAIL Q26–Q30**

30 Jun 2026 | ANSWER KEY, ★ method & the trap — for the parent

Q26. [M2] Answer: 234 [M2 · stacked-constraint number · generated twin of 2025 Q26 · generated · 6 marks]

What is the smallest three-digit number with all of these properties: divisible by 6; contains the digit 3; has no digit 0; has all different digits; has exactly 1 odd digit?

★ Method: Apply the strong filters first (divisibility, no-zero, all-different), THEN check the subtle last condition before you commit — don't stop at the first multiple you find.

The trap: a solver who skips "has exactly 1 odd digit" stops at 132 — the planted decoy (a real answer to a weaker question).

Q27. [M5] Answer: 448 [M5 · organised counting (residue mod) · generated twin of 2013 Q27 · generated · 7 marks]

In how many ways can you choose 3 different numbers from 2 to 22 so that their sum is a multiple of 3?

★ Method: Bucket every number by remainder mod 3; valid triples are 'all three from one bucket' PLUS the matching one-from-each cross-bucket combinations — add the $C(\cdot, 3)$ and product counts.

The trap: $C(21, 3) = 1330$ triples — listing them is impossible in time, and 'choose' is unordered so (a,b,c) must not be recounted as (b,a,c).

Q28. [M6] Answer: 37 [M6 · iterate a map → find the cycle · generated twin of 2024 Q29 · generated · 8 marks]

Start with the number 8665. At each step replace the number with the sum of the squares of its digits (for example $24 \rightarrow 2^2 + 4^2 = 20$). What is the 3996th number in this list?

★ Method: Iterate until the numbers REPEAT, then you have a cycle. Find the cycle length L and where it starts; the 3996th term = step into the cycle by $(3996 - \text{start}) \bmod L$. Never iterate 3996 times.

The trap: an off-by-one in where the cycle starts, or not noticing it repeats.

Q29. [M7] Answer: 30 [M7 · area dissection (exact figure) · generated twin of 2024 Q23 · figure (coordinate-exact) · 9 marks]

ABCD is a rectangle measuring 8 by 10 (area 80). M is the midpoint of side BC and N is the midpoint of side DC. M and N are each joined to A and to each other. What is the area of the shaded triangle AMN?

★ Method: Don't trust the not-to-scale picture. Put the rectangle on a grid and subtract the THREE corner right-triangles from the whole rectangle — no area formula needed. (Closed form here is $3 \times W \times H / 8$.)

The trap: Reading lengths off the drawing instead of the stated dimensions; the figure is deliberately not to scale.

Q30. [M8] Answer: 636 [M8 · cube opposite-faces invariant · generated twin of 2018 Q27 · generated · 10 marks]

Using only the digits 1, 2, 4, each face of a cube shows a different 3-digit number. The three faces meeting at one corner show 214, 222 and 224. The two numbers on each pair of opposite faces add to the same total. What is the largest that this total could be?

★ Method: The invariant: all three opposite-face pairs share ONE common sum S. The three shown faces meet at a corner so they are mutually NON-opposite — their three partners are the remaining numbers. Pick the pairing that maximises S, then verify all three pairs really hit the same S.

The trap: assuming the three visible faces can be opposite each other — they meet at a corner, so they are adjacent; their opposites are the three hidden faces.

Coverage: M2 stacked-constraint number · M5 organised counting · M6 repeating-map cycle (invariant, not counting) · M7 area dissection (coordinate-exact figure) · M8 cube opposite-faces invariant. Five mechanisms, five methods — the balanced prize tail.